

# DANIEL ADEJUMO

+1 (917) 664-8939 | [dxa239@nyu.edu](mailto:dxa239@nyu.edu) | Brooklyn, New York  
[linkedin.com/in/daniel-adejumo](https://www.linkedin.com/in/daniel-adejumo) | [github.com/danieladejumo17](https://github.com/danieladejumo17)

## EDUCATION

### New York University

New York, USA

**M.S. in Mechatronics and Robotics Engineering (CGPA: 3.54/4.0)**

**September 2024 - May 2026**

Relevant Coursework: Reinforcement Learning & Optimal Control, Mathematics for Robotics, Advanced Mechatronics

### University of Ibadan

Ibadan, Nigeria

**B.Sc. Mechanical Engineering (First Class Honors, CGPA 3.70/4.0)**

**June 2023**

B.Sc. Thesis: UI Hand I - The Modelling, Design, and Control of a Dexterous Robotic Hand

## RESEARCH INTEREST

Dynamics and Control, First-Order Model-Based Reinforcement Learning and constrained RL for robust and safe control of high-dimensional robotic systems.

## PUBLICATIONS

**Adejumo, D.O.**, Fadare, D.A., Kazeem, R.A., Ikumapayi, O.M., Falana, A., Adedayo, A.S., Fadare, D.A., Adeoye, A.O.M., Ogundipe, A.T., Olarinde, E.S. (2023). A low-cost, modular, cable-driven, anthropomorphic robotic hand: A conceptual design and application in biomimetic study. *Journal Européen des Systèmes Automatisés*, Vol. 56, No. 4, pp. 627-639. <https://doi.org/10.18280/jesa.560412>

**Adejumo D. O.**, Fadare, D. A., Falana, A., Fadare, D. A. (2022). UI Hand I: The Conceptual Design for a Low-Cost and Modular 26-DOF Anthropomorphic Robotic Hand. In 2022 35th International Conference and Annual General Meeting, The Nigerian Institution of Mechanical Engineers (pp. 119 –134).

[Manuscript] Autonomous Mobile Drawing Robot

## SKILLS

- **Programming and Software:** Python, C/C++, MATLAB, JavaScript, OpenCV, Git, Linux, Docker
- **Machine Learning:** Deep Learning, Deep Reinforcement Learning, LLMs, VLMs, PyTorch, HuggingFace
- **Robotics and Embedded Systems:** Electronics, Microcontroller Programming, Sensor/Actuator Interfacing, PID and MPC Control, Robot Operating System (ROS), Gazebo, Simscape, MuJoCo, IsaacSim & Isaac Lab
- **Design (CAD) and Rapid Prototyping:** Fusion 360, SolidWorks, AutoCAD, 3D Printing, CNC Machining
- **Soft Skills:** Communication, Teamwork, Project Management, Problem-solving

## PROJECTS

### **Anomaly Detection for Autonomous Vehicles**

**September 2025 – December 2025**

- Built a video-based anomaly detection pipeline for Autonomous Vehicles (AVs) using Nvidia Cosmos-Reason1 (Qwen2.5-VL-7B-Instruct) Vision-Language Model (VLM).
- Optimized the inference pipeline and applied prompt-tuning and pruning, reducing reasoning time by 80%.
- Implemented FP4 quantization for deployment on an RTX 5090, achieving ~500 ms end-to-end reasoning for real-time video anomaly detection while maintaining accuracy.

### **Reward Shaping for Quadruped Locomotion**

**November 2025 – December 2025**

- Trained a command-conditioned quadrupedal locomotion policy using PPO for the Unitree Go2 robot in IsaacLab, leveraging NYU's high-performance computing (HPC) infrastructure.
- Designed and optimized reward functions to produce stable, natural-looking, and robust gaits; extended the same framework to enable backflip with recovery and bipedal (hind-leg) locomotion on the Go2.
- Applied domain randomization across contact friction, inertial properties, and actuator dynamics to improve robustness and reduce the sim-to-real gap.

### **Reinforcement Learning for Gait Control in Bipedal Robots**

**January 2025 – April 2025**

- Trained velocity-conditioned locomotion controllers for Unitree H1 and G1 bipedal robots in the Genesis physics engine using reinforcement learning (PPO), enabling dynamic motion control.
- Applied advanced reward engineering techniques to improve gait stability and policy robustness.
- Initiated Sim2Real policy transfer on a Unitree G1 to evaluate policy generalization to the real world.

### **Design and Control of an Anthropomorphic Robotic Hand (B.Sc. Thesis)**

**February 2022 – June 2023**

- Designed and prototyped a biomimetic robotic hand, integrating tendon-driven actuation and embedded sensing to enable dexterous manipulation tasks for industrial automation scenarios.

- Engineered the embedded control architecture, including actuator integration, microcontroller communication protocols, and low-level motor control using C++ and Python on Raspberry Pi and Pico MCUs.
- Prepared and reviewed an academic paper, including a comprehensive literature review, collaborated with co-authors, and published the paper in the Journal Européen des Systèmes Automatisés.

#### **Autonomous Mobile Drawing Robot**

**January 2025 – April 2025**

- Designed a mobile robotic system with an integrated pen mechanism to overcome workspace limitations of fixed-base automated drawing platforms, utilizing Arduino and Propeller microcontrollers for control.
- Developed an image-to-G-code processing pipeline in C++ and Python, using computer vision techniques (OpenCV) to convert raster images, SVG, and DXF formats into the mobile robot's motion commands.
- Designed detailed CAD models in Fusion 360 and fabricated the system with 3D printing.

#### **Course Projects (Foundation of Robotics and Mechatronics)**

**September 2024 – December 2024**

- Implemented forward and inverse kinematics for a 6-DOF robotic arm in MATLAB, solving nonlinear equations and validating solutions through MATLAB-based visualizations.
- Designed and integrated a mobile robotic system with an articulated arm for autonomous detection and removal of liquid spills and trip hazards in household environments.
- Repurposed a capacitive moisture sensor as a spill detector by leveraging capillary action in foam materials.

### **EXPERIENCE**

#### **New York University**

New York, USA

##### **Graduate Teaching Assistant (Advanced Mechatronics)**

**January 2026 – Present**

- Advising students on course content and questions during office hours
- Providing hands-on technical support for students' weekly mini-projects and semester projects.

##### **Course Assistant (Innovation & Science of Smart Cities)**

**July 2025 – August 2025**

- Prepared a detailed 5-week program and class activities for the Innovation, Entrepreneurship and Science of Smart Cities (IESoSC) summer program at the NYU Center for K12 STEM Education.
- Delivered class instructions to 25 students, supported students with hands-on class activities, and supervised students' projects to prepare them for the Shark Tank-style showcase at the end of the program.
- Collaborated with other class instructors, managed students' data, and communicated class progress and requirements to program managers to ensure a seamless and successful program.

#### **Soft Quest Incorporated (SQI) College of ICT**

Ibadan, Nigeria

##### **Automation and Robotic Systems Engineer**

**September 2022 – August 2024**

- Trained students to acquire foundational skills in robotics, including electronics design, embedded systems, programming, Computer-Aided Design (CAD), and rapid prototyping (3D printing, CNC machining).
- Collaborated with other teams and departments at SQI to organize events, including a 1-week robotics training bootcamp welcoming over 100 students from nearby institutions.
- Installed and maintained equipment, including 3D Printers, CNC Laser Cutting Machines, and PCB Printers.

##### **Control Systems & Simulation Intern, Robotics Department**

**August 2021 – February 2022**

- Developed and validated the powertrain subsystem for the Alpha Electric Vehicle using model-based design in MATLAB and Simulink, applying control system principles to optimize dynamic performance and accelerate design iteration cycles.
- Delivered design insights to technical and non-technical stakeholders through structured presentations and simulation-driven visuals, supporting clear decision-making throughout the project lifecycle.
- Designed and prototyped embedded automation systems, including a gas metering system and security alert system, using sensor integration and microcontroller logic.

#### **Nigerian Institution of Mechanical Engineers, University of Ibadan**

Ibadan, Nigeria

##### **Team Lead, Control Electronics Subsystem – Project Helios**

**December 2019 – May 2020**

- Led a 6-member team to develop the electronics and control subsystems for a student electric vehicle project, overseeing system design, component selection, and integration strategy.
- Defined design requirements for the powertrain and electrical systems, ensuring alignment with control objectives and overall vehicle architecture.

### **AWARDS**

- Students' Choice Award, NYU Tandon Capstone Project Competition (Team Award, April 2025)
- SEED Scholarship, New York University Tandon School of Engineering (2024 – 2026)
- Dean's Honor Awards, Faculty of Technology, University of Ibadan (5x, 2017 – 2023)
- Outstanding Cambridge Learner Award, Top in Nigeria, Physics (November 2016)